

SOLIDWORKS API (NET VBA C#) — Reference (203/337): IModeler (part 1 of 4)

SOLIDWORKS API Documentation (converted from CHM)

IModeler (part 1 of 4) (Type)

IModeler Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModeler Interface

Provides an interface to temporary body objects.

NOTE: Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

.NET Syntax

Visual Basic (Declaration)

```
Public Interface IModeler
```

Visual Basic (Usage)



```
Dim instance As IModeler
```

C#

```
public interface IModeler
```

C++/CLI

```
public interface class IModeler
```

Visual Basic for Applications (VBA) Syntax

See [Modeler](#).

Example

[Check Interference Using IModeler::CheckInterference \(VBA\)](#),
[Create Loft Body \(VB.NET\)](#),
[Create Loft Body \(VBA\)](#),
[Create Loft Body \(C#\)](#)

Accessors

[ISldWorks::Get Modeler](#)

Access Diagram

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See Also

[IModeler Members](#)
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IModeler Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModeler Interface

For a list of all members of this type, see [IModeler members](#).

Public Methods

Name	Description
 CheckInterference	Obsolete. Superseded by IModeler::CheckInterference3 .
 CheckInterference3	Checks for interferences among the specified bodies in a part.

	CheckInterferenceBetweenTwoBodies	Checks for interference between the specified bodies in an assembly.
	CopyWizardHole	Copies hole data from the source hole to the destination hole.
	CreateArc	Creates an arc.
	CreateBodiesFromSheets	Obsolete. Superseded by IModeler.CreateBodiesFromSheets2
	CreateBodiesFromSheets2	Sews sheets to make sheet (surface) or solid bodies.
	CreateBodyFromBox	Obsolete. Superseded by IModeler.CreateBodyFromBox3
	CreateBodyFromBox3	Creates a temporary body from the specified box dimensions.
	CreateBodyFromCone	Creates a temporary body from cone dimensions.
	CreateBodyFromCyl	Creates a temporary body from cylinder dimensions.
	CreateBodyFromFaces2	Creates a temporary body from a list of faces.
	CreateBrepBody	Obsolete. Superseded by IModeler.CreateBrepBody3
	CreateBrepBody3	Creates a temporary body from BREP (boundary representation) data.
	CreateBsplineCurve	Creates a b-spline curve.
	CreateBsplineSurface	Creates a b-spline surface.
	CreateConicalSurface	Obsolete. Superseded by IModeler.CreateConicalSurface2
	CreateConicalSurface2	Creates an untrimmed conical surface.
	CreateCoonsBSurface	Creates a coons b-surface using the four specified boundary curves.
	CreateCylindricalSurface	Obsolete. Superseded by IModeler.CreateCylindricalSurface2
	CreateCylindricalSurface2	Creates an untrimmed cylindrical surface.
	CreateEllipse	Creates a temporary elliptical curve.
	CreateExtrudedBody	Creates a temporary extruded body.
	CreateExtrusionSurface	Creates an extruded surface.
	CreateLine	Creates a line.
	CreateLoftBody	Obsolete. Superseded by IModeler.CreateLoftBody2
	CreateLoftBody2	Creates a loft body using the specified profiles, guide curves, and centerline.
	CreateLoftSurface	Creates a loft surface.
	CreateOffsetSurface	Creates a surface that is offset from an existing surface.
	CreatePCurve	Creates a pcurve.
	CreatePlanarSurface	Obsolete. Superseded by IModeler.CreatePlanarSurface2
	CreatePlanarSurface2	Creates a new infinite planar surface.
	CreateRuledSurface	Creates a ruled surface from the curves.
	CreateRuledSurfaceFromEdges	Creates a ruled surface using the specified edges and returns a surface body.
	CreateSheetFromFaces	Creates a sheet (surface) body from connected faces.
	CreateSheetFromSurface	Creates a sheet (surface) body from a surface.
	CreateSphericalSurface	Obsolete. Superseded by IModeler.CreateSphericalSurface2
	CreateSphericalSurface2	Creates an untrimmed spherical surface.
	CreateSpring	Creates the geometry for a spring.
	CreateSweptBody	Creates a swept body.
	CreateSweptSurface	Creates a swept surface from a curve.
	CreateToroidalSurface	Creates an untrimmed toroidal surface from the specified arguments.
	CreateWireBody	Creates a wire body from the input entities, which can be edges or curves.
	DeleteFacesFromSheetBody	Deletes the unused faces of the sheet body.
	FindTwoEdgeMaxDeviation	Finds the maximum distance between two edges.

		
	GetBodyOutline	Obsolete. Superseded by IModeler.GetBodyOutline2 .
	GetBodyOutline2	Gets the curves that form the outline of a body.
	GetInitKnitGapWidth	Gets the initial gap bound width for knitting a body.
	GetManifoldBodiesCount	Gets the number of manifold bodies created from the specified non-manifold body.
	GetToleranceValue	Gets the current tolerance value of the given tolerance ID.
	ICheckInterference	Obsolete. Superseded by IModeler.ICheckInterference2 .
	ICheckInterference2	Obsolete. Superseded by IModeler.ICheckInterference3 .
	ICheckInterference3	Checks the interference among the specified bodies.
	ICheckInterferenceCount	Obsolete. Superseded by IModeler.ICheckInterferenceCount2 .
	ICheckInterferenceCount2	Obsolete. Superseded by IModeler.ICheckInterferenceCount3 .
	ICheckInterferenceCount3	Checks interference among the specified bodies and returns the number of interferences.
	ICopyWizardHole	Copies hole data from the source hole to the destination hole.
	ICreateArc	Creates an arc.
	ICreateBodiesFromSheets	Obsolete. Superseded by IModeler.ICreateBodiesFromSheets2 .
	ICreateBodiesFromSheets2	Sews sheets to make a sheet (surface) or solid bodies.
	ICreateBodyFromBox	Obsolete. Superseded by IModeler.ICreateBodyFromBox2 .
	ICreateBodyFromBox2	Obsolete. Superseded by IModeler.ICreateBodyFromBox3 .
	ICreateBodyFromCone	Obsolete. Superseded by IModeler.ICreateBodyFromCone2 .
	ICreateBodyFromCone2	Creates a temporary body from cone dimensions.
	ICreateBodyFromCyl	Obsolete. Superseded by IModeler.ICreateBodyFromCyl2 .
	ICreateBodyFromCyl2	Creates a temporary body from cylinder dimensions.
	ICreateBodyFromFaces	Obsolete. Superseded by IModeler.ICreateBodyFromFace3 .
	ICreateBodyFromFaces2	Obsolete. Superseded by IModeler.ICreateBodyFromFace3 .
	ICreateBodyFromFaces3	Creates a temporary body from a list of faces.
	ICreateBrepBody	Obsolete. Superseded by IModeler.ICreateBrepBody3 .
	ICreateBrepBody2	Obsolete. Superseded by IModeler.ICreateBrepBody3 .
	ICreateBrepBody3	Creates a temporary body from BREP (boundary representation) data.
	ICreateBsplineCurve	Creates a b-spline curve.
	ICreateBsplineSurface	Creates a b-spline surface.
	ICreateConicalSurface	Obsolete. Superseded by IModeler.ICreateConicalSurface2 .
	ICreateConicalSurface2	Creates an untrimmed conical surface.
	ICreateCylindricalSurface	Obsolete. Superseded by IModeler.ICreateCylindricalSurface2 .
	ICreateCylindricalSurface2	Creates an untrimmed cylindrical surface.
	ICreateEllipse	Creates a temporary elliptical curve.
	ICreateExtrusionSurface	Creates an extruded surface.
	ICreateLine	Creates a line.
	ICreateLoftSurface	Creates a loft surface.
	ICreateOffsetSurface	Creates a surface that is offset from an existing surface.
	ICreatePCurve	Creates a pcurve.
	ICreatePlanarSurface	Obsolete. Superseded by IModeler.ICreatePlanarSurface2 .
	ICreatePlanarSurface2	Creates a new infinite planar surface.
	ICreateRuledSurface	Creates a ruled surface from the curves.

		
	ICreateRuledSurfaceFromEdge	Creates a ruled surface using the specified edges and returns a surface body.
	ICreateSheetFromFaces	Creates a sheet (surface) body from connected faces.
	ICreateSheetFromSurface	Obsolete. Superseded by IModeler.ICreateSheetFromSurface2
	ICreateSheetFromSurface2	Creates a sheet (surface) body from a surface.
	ICreateSphericalSurface	Obsolete. Superseded by IModeler.ICreateSphericalSurface2 .
	ICreateSphericalSurface2	Creates an untrimmed spherical surface.
	ICreateSweptSurface	Creates a swept surface from a curve.
	ICreateToroidalSurface	Creates an untrimmed toroidal surface from the specified arguments.
	ICreateWireBody	Creates a wire body from the input entities, which can be edges or curves.
	IDeleteFacesFromSheetBody	Deletes the unused faces of the sheet body.
	IFindTwoEdgeMaxDeviation	Finds the maximum distance between two edges.
	IImprintingFaces	Imprints the specified tool faces onto the specified target faces.
	IImprintingFacesCount	Obsolete. Superseded by IModeler.IImprintingFacesCount2 .
	IImprintingFacesCount2	Gets the number of imprinted edges and vertices in the model.
	IMakeManifoldBodies	Creates manifold bodies from the specified non-manifold body.
	IMergeCurves	Merges multiple curves into one curve.
	IImprintingFaces	Imprints the specified tool faces onto the specified target faces.
	IReplaceSurfaces	Obsolete. Superseded by IModeler.IReplaceSurfaces2 .
	IReplaceSurfaces2	Replaces the surfaces of the given faces.
	IRestore	Obsolete. Superseded by IModeler.IRestore2 .
	IRestore2	Restores a temporary body object from the specified stream.
	ISplitFaceOnParam	Obsolete. Superseded by IModeler.ISplitFaceOnParam2 .
	ISplitFaceOnParam2	Splits and retrieves the faces on the U or V parameter.
	ISplitFaceOnParamCount	Obsolete. Superseded by IModeler.ISplitFaceOnParamCount2 .
	ISplitFaceOnParamCount2	Sets up and counts the number of new faces split on the U or V parameter.
	MakeManifoldBodies	Creates manifold bodies from the specified non-manifold body.
	MergeCurves	Merges multiple curves into one curve.
	ProjectCurveOnSurface	Projects a curve on a surface.
	ReplaceSurfaces	Replaces the surfaces of the given faces.
	Restore	Restores a temporary body object from the specified stream.
	SetInitKnitGapWidth	Sets the initial gap bound width for sewing a body. Call this method before calling any calls that attempt to knit a body; for example, IBody2.CreateBodyFromSurfaces .
	SetTolerances	Obsolete. Superseded by IModeler.GetToleranceValue and IModeler.SetToleranceValue .
	SetToleranceValue	Sets tolerances governing geometry output.
	SplitFaceOnParam	Splits and retrieves the faces on the U or V parameter
	ThickenSheet	Thickens a sheet body.
	UnsetTolerances	Sets the tolerances back to system settings.

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See Also

[IModeler Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

[See Also](#) [Properties](#) [Methods](#)

 [SolidWorks.Interop.sldworks](#) Namespace : IModeler Interface

The following tables list the members exposed by [IModeler](#).

Public Properties

Name	Description
 GeneralTopology	Gets or sets the Parasolid option to create general (non-manifold) bodies.

[Top](#)

Public Methods

Name	Description
 CheckInterference	Obsolete. Superseded by IModeler.CheckInterference3 .
 CheckInterference3	Checks for interferences among the specified bodies in a part.
 CheckInterferenceBetweenTwoBodies	Checks for interference between the specified bodies in an assembly.
 CopyWizardHole	Copies hole data from the source hole to the destination hole.
 CreateArc	Creates an arc.
 CreateBodiesFromSheets	Obsolete. Superseded by IModeler.CreateBodiesFromSheets2 .
 CreateBodiesFromSheets2	Sews sheets to make sheet (surface) or solid bodies.
 CreateBodyFromBox	Obsolete. Superseded by IModeler.CreateBodyFromBox3 .
 CreateBodyFromBox3	Creates a temporary body from the specified box dimensions.
 CreateBodyFromCone	Creates a temporary body from cone dimensions.
 CreateBodyFromCyl	Creates a temporary body from cylinder dimensions.
 CreateBodyFromFaces2	Creates a temporary body from a list of faces.
 CreateBrepBody	Obsolete. Superseded by IModeler.CreateBrepBody3 .
 CreateBrepBody3	Creates a temporary body from BREP (boundary representation) data.
 CreateB splineCurve	Creates a b-spline curve.
 CreateB splineSurface	Creates a b-spline surface.
 CreateConicalSurface	Obsolete. Superseded by IModeler.CreateConicalSurface2 .
 CreateConicalSurface2	Creates an untrimmed conical surface.
 CreateCoonsBSurface	Creates a coons b-surface using the four specified boundary curves.
 CreateCylindricalSurface	Obsolete. Superseded by IModeler.CreateCylindricalSurface2 .
 CreateCylindricalSurface2	Creates an untrimmed cylindrical surface.
 CreateEllipse	Creates a temporary elliptical curve.
 CreateExtrudedBody	Creates a temporary extruded body.
 CreateExtrusionSurface	Creates an extruded surface.
 CreateLine	Creates a line.
 CreateLoftBody	Obsolete. Superseded by IModeler.CreateLoftBody2 .
 CreateLoftBody2	Creates a loft body using the specified profiles, guide curves, and centerline.
 CreateLoftSurface	Creates a loft surface.
 CreateOffsetSurface	Creates a surface that is offset from an existing surface.
 CreatePCurve	Creates a pcurve.
 CreatePlanarSurface	Obsolete. Superseded by IModeler.CreatePlanarSurface2 .
 CreatePlanarSurface2	Creates a new infinite planar surface.

	CreateRuledSurface	Creates a ruled surface from the curves.
	CreateRuledSurfaceFromEdge	Creates a ruled surface using the specified edges and returns a surface body.
	CreateSheetFromFaces	Creates a sheet (surface) body from connected faces.
	CreateSheetFromSurface	Creates a sheet (surface) body from a surface.
	CreateSphericalSurface	Obsolete. Superseded by IModeler.CreateSphericalSurface2
	CreateSphericalSurface2	Creates an untrimmed spherical surface.
	CreateSpring	Creates the geometry for a spring.
	CreateSweptBody	Creates a swept body.
	CreateSweptSurface	Creates a swept surface from a curve.
	CreateToroidalSurface	Creates an untrimmed toroidal surface from the specified arguments.
	CreateWireBody	Creates a wire body from the input entities, which can be edges or curves.
	DeleteFacesFromSheetBody	Deletes the unused faces of the sheet body.
	FindTwoEdgeMaxDeviation	Finds the maximum distance between two edges.
	GetBodyOutline	Obsolete. Superseded by IModeler.GetBodyOutline2
	GetBodyOutline2	Gets the curves that form the outline of a body.
	GetInitKnitGapWidth	Gets the initial gap bound width for knitting a body.
	GetManifoldBodiesCount	Gets the number of manifold bodies created from the specified non-manifold body.
	GetToleranceValue	Gets the current tolerance value of the given tolerance ID.
	ICheckInterference	Obsolete. Superseded by IModeler.ICheckInterference2
	ICheckInterference2	Obsolete. Superseded by IModeler.ICheckInterference3
	ICheckInterference3	Checks the interference among the specified bodies.
	ICheckInterferenceCount	Obsolete. Superseded by IModeler.ICheckInterferenceCount2
	ICheckInterferenceCount2	Obsolete. Superseded by IModeler.ICheckInterferenceCount3
	ICheckInterferenceCount3	Checks interference among the specified bodies and returns the number of interferences.
	ICopyWizardHole	Copies hole data from the source hole to the destination hole.
	ICreateArc	Creates an arc.
	ICreateBodiesFromSheets	Obsolete. Superseded by IModeler.ICreateBodiesFromSheets2
	ICreateBodiesFromSheets2	Sews sheets to make a sheet (surface) or solid bodies.
	ICreateBodyFromBox	Obsolete. Superseded by IModeler.ICreateBodyFromBox2
	ICreateBodyFromBox2	Obsolete. Superseded by IModeler.CreateBodyFromBox3
	ICreateBodyFromCone	Obsolete. Superseded by IModeler.ICreateBodyFromCone2
	ICreateBodyFromCone2	Creates a temporary body from cone dimensions.
	ICreateBodyFromCyl	Obsolete. Superseded by IModeler.ICreateBodyFromCyl2
	ICreateBodyFromCyl2	Creates a temporary body from cylinder dimensions.
	ICreateBodyFromFaces	Obsolete. Superseded by IModeler.ICreateBodyFromFace3
	ICreateBodyFromFaces2	Obsolete. Superseded by IModeler.ICreateBodyFromFace3
	ICreateBodyFromFaces3	Creates a temporary body from a list of faces.
	ICreateBrepBody	Obsolete. Superseded by IModeler.ICreateBrepBody3
	ICreateBrepBody2	Obsolete. Superseded by IModeler.ICreateBrepBody3
	ICreateBrepBody3	Creates a temporary body from BREP (boundary representation) data.
	ICreateBsplineCurve	Creates a b-spline curve.
	ICreateBsplineSurface	Creates a b-spline surface.
	ICreateConicalSurface	Obsolete. Superseded by IModeler.ICreateConicalSurface2

		
	ICreateConicalSurface2	Creates an untrimmed conical surface.
	ICreateCylindricalSurface	Obsolete. Superseded by IModeler:ICreateCylindricalSurface2 .
	ICreateCylindricalSurface2	Creates an untrimmed cylindrical surface.
	ICreateEllipse	Creates a temporary elliptical curve.
	ICreateExtrusionSurface	Creates an extruded surface.
	ICreateLine	Creates a line.
	ICreateLoftSurface	Creates a loft surface.
	ICreateOffsetSurface	Creates a surface that is offset from an existing surface.
	ICreatePCurve	Creates a pcurve.
	ICreatePlanarSurface	Obsolete. Superseded by IModeler:ICreatePlanarSurface2 .
	ICreatePlanarSurface2	Creates a new infinite planar surface.
	ICreateRuledSurface	Creates a ruled surface from the curves.
	ICreateRuledSurfaceFromEdge	Creates a ruled surface using the specified edges and returns a surface body.
	ICreateSheetFromFaces	Creates a sheet (surface) body from connected faces.
	ICreateSheetFromSurface	Obsolete. Superseded by IModeler:ICreateSheetFromSurface2 .
	ICreateSheetFromSurface2	Creates a sheet (surface) body from a surface.
	ICreateSphericalSurface	Obsolete. Superseded by IModeler:ICreateSphericalSurface2 .
	ICreateSphericalSurface2	Creates an untrimmed spherical surface.
	ICreateSweptSurface	Creates a swept surface from a curve.
	ICreateToroidalSurface	Creates an untrimmed toroidal surface from the specified arguments.
	ICreateWireBody	Creates a wire body from the input entities, which can be edges or curves.
	IDeleteFacesFromSheetBody	Deletes the unused faces of the sheet body.
	IFindTwoEdgeMaxDeviation	Finds the maximum distance between two edges.
	IImprintingFaces	Imprints the specified tool faces onto the specified target faces.
	IImprintingFacesCount	Obsolete. Superseded by IModeler:IImprintingFacesCount2 .
	IImprintingFacesCount2	Gets the number of imprinted edges and vertices in the model.
	IMakeManifoldBodies	Creates manifold bodies from the specified non-manifold body.
	IMergeCurves	Merges multiple curves into one curve.
	ImprintingFaces	Imprints the specified tool faces onto the specified target faces.
	IReplaceSurfaces	Obsolete. Superseded by IModeler:IReplaceSurfaces2 .
	IReplaceSurfaces2	Replaces the surfaces of the given faces.
	IRestore	Obsolete. Superseded by IModeler:IRestore2 .
	IRestore2	Restores a temporary body object from the specified stream.
	ISplitFaceOnParam	Obsolete. Superseded by IModeler:ISplitFaceOnParam2 .
	ISplitFaceOnParam2	Splits and retrieves the faces on the U or V parameter.
	ISplitFaceOnParamCount	Obsolete. Superseded by IModeler:ISplitFaceOnParamCount2 .
	ISplitFaceOnParamCount2	Sets up and counts the number of new faces split on the U or V parameter.
	MakeManifoldBodies	Creates manifold bodies from the specified non-manifold body.
	MergeCurves	Merges multiple curves into one curve.
	ProjectCurveOnSurface	Projects a curve on a surface.
	ReplaceSurfaces	Replaces the surfaces of the given faces.
	Restore	Restores a temporary body object from the specified stream.

		
	SetInitKnitGapWidth	Sets the initial gap bound width for sewing a body. Call this method before calling any calls that attempt to knit a body; for example, IBody2.CreateBodyFromSurfaces .
	SetTolerances	Obsolete. Superseded by IModeler.GetToleranceValue and IModeler.SetToleranceValue .
	SetToleranceValue	Sets tolerances governing geometry output.
	SplitFaceOnParam	Splits and retrieves the faces on the U or V parameter
	ThickenSheet	Thickens a sheet body.
	UnsetTolerances	Sets the tolerances back to system settings.

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See Also

[IModeler Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

IModeler Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IModeler Interface

For a list of all members of this type, see [IModeler members](#).

Public Properties

Name	Description
 GeneralTopology	Gets or sets the Parasolid option to create general (non-manifold) bodies.

[Top](#)

See Also

[IModeler Interface](#)
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

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CheckInterference Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CheckInterference Method (IModeler)

Obsolete. Superseded by [IModeler::CheckInterference3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CheckInterference( _  
    ByVal Body1 As System.Object, _  
    ByVal Body2 As System.Object, _  
    ByVal CoincidentInterference As System.Boolean, _  
    ByRef Body1InterferedFaceArray As System.Object, _  
    ByRef Body2InterferedFaceArray As System.Object, _  
    ByRef IntersectedBodyArray As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1 As System.Object  
Dim Body2 As System.Object  
Dim CoincidentInterference As System.Boolean  
Dim Body1InterferedFaceArray As System.Object  
Dim Body2InterferedFaceArray As System.Object  
Dim IntersectedBodyArray As System.Object  
Dim value As System.Boolean
```

```
value = instance.CheckInterference(Body1, Body2, CoincidentInterference, Body1InterferedFaceArray, Body2InterferedFaceArray, IntersectedBodyArray)
```

C#

```
System.bool CheckInterference(  
    System.object Body1,  
    System.object Body2,  
    System.bool CoincidentInterference,  
    out System.object Body1InterferedFaceArray,  
    out System.object Body2InterferedFaceArray,  
    out System.object IntersectedBodyArray  
)
```

C++/CLI

```
System.bool CheckInterference(  
    & System.Object^ Body1,  
    & System.Object^ Body2,  
    & System.bool CoincidentInterference,  
    [Out] System.Object^ Body1InterferedFaceArray,  
    [Out] System.Object^ Body2InterferedFaceArray,  
    [Out] System.Object^ IntersectedBodyArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body1

First [body](#) to check for interference

Body2

Second [body](#) to check for interference

CoincidentInterference

True to check for coincident interference, false to not

Body1InterferedFaceArray

Array of [faces](#) that have interfered from the first body with the second body

Body2InterferedFaceArray

Array of [faces](#) that have interfered from the second body with the first body

IntersectedBodyArray

Array of the interfering [bodies](#)

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::CheckInterference](#).

Example

[Check Interference Using IModeler::CheckInterference \(VBA\)](#)

Remarks

Each body must be transformed in the coordinate space of the top-level assembly so that it is positioned the same with respect to the other bodies as it is in assembly space.

To align the two bodies, apply the transformation from [IComponent2::Transform2](#) using [IBody2::ApplyTransform](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::ICheckInterferenceCount2 Method](#)

[IAssemblyDoc::IToolsCheckInterference3 Method](#)

[IAssemblyDoc::ToolsCheckInterference2 Method](#)

Availability

CheckInterference3 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CheckInterference3 Method (IModeler)

Checks for interferences among the specified bodies in a part.

.NET Syntax

Visual Basic (Declaration)

```
Function CheckInterference3( _
    ByVal TargetBodies As System.Object, _
    ByVal ToolBodies As System.Object, _
    ByVal Options As System.Integer, _
    ByRef Body1InterferedFaceArray As System.Object, _
    ByRef Body2InterferedFaceArray As System.Object, _
    ByRef IntersectedBodyArray As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim TargetBodies As System.Object
Dim ToolBodies As System.Object
Dim Options As System.Integer
Dim Body1InterferedFaceArray As System.Object
Dim Body2InterferedFaceArray As System.Object
Dim IntersectedBodyArray As System.Object
Dim value As System.Boolean
```

```
value = instance.CheckInterference3(TargetBodies, ToolBodies, Options, Body1InterferedFaceArray,
Body2InterferedFaceArray, IntersectedBodyArray)
```

C#

```
System.bool CheckInterference3(
    System.object TargetBodies,
    System.object ToolBodies,
    System.int Options,
    out System.object Body1InterferedFaceArray,
    out System.object Body2InterferedFaceArray,
    out System.object IntersectedBodyArray
)
```

C++/CLI

```
System.bool CheckInterference3(
    & System.Object^ TargetBodies,
    & System.Object^ ToolBodies,
    & System.int Options,
    [Out] System.Object^ Body1InterferedFaceArray,
    [Out] System.Object^ Body2InterferedFaceArray,
    [Out] System.Object^ IntersectedBodyArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

TargetBodies

Target [bodies](#)

ToolBodies

Tool [bodies](#)

Options

Check interference options as defined by [swCheckInterferenceOption_e](#)

Body1InterferedFaceArray

Array of [faces](#) that interfered from the first body with the second body

Body2InterferedFaceArray

Array of [faces](#) that interfered from the second body with the first body

IntersectedBodyArray

Array of interfering [bodies](#)

Return Value

True if an interference exists, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::CheckInterference3](#).

Example

[Check Interference Among Bodies \(VBA\)](#)

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ICheckInterference3 Method](#)
[IModeler::ICheckInterference3 Method](#)
[IAssemblyDoc::IToolsCheckInterference3 Method](#)
[IAssemblyDoc::ToolsCheckInterference2 Method](#)
[InterferenceDetectionMgr Interface](#)
[IModeler::CheckInterferenceBetweenTwoBodies Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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CheckInterferenceBetweenTwoBodies Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CheckInterferenceBetweenTwoBodies Method (IModeler)

Checks for interference between the specified bodies in an assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function CheckInterferenceBetweenTwoBodies( _  
    ByVal Body1 As System.Object, _  
    ByVal Body2 As System.Object, _  
    ByVal CoincidentInterference As System.Boolean, _  
    ByRef Body1InterferedFaceArray As System.Object, _  
    ByRef Body2InterferedFaceArray As System.Object, _  
    ByRef IntersectedBodyArray As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Body1 As System.Object  
Dim Body2 As System.Object  
Dim CoincidentInterference As System.Boolean  
Dim Body1InterferedFaceArray As System.Object  
Dim Body2InterferedFaceArray As System.Object  
Dim IntersectedBodyArray As System.Object  
Dim value As System.Boolean  
  
value = instance.CheckInterferenceBetweenTwoBodies(Body1, Body2, CoincidentInterference,  
Body1InterferedFaceArray, Body2InterferedFaceArray, IntersectedBodyArray)
```

C#

```
System.bool CheckInterferenceBetweenTwoBodies(
    System.object Body1,
    System.object Body2,
    System.bool CoincidentInterference,
    out System.object Body1InterferedFaceArray,
    out System.object Body2InterferedFaceArray,
    out System.object IntersectedBodyArray
)
```

C++/CLI

```
System.bool CheckInterferenceBetweenTwoBodies(
    & System.Object^ Body1,
    & System.Object^ Body2,
    & System.bool CoincidentInterference,
    [Out] System.Object^ Body1InterferedFaceArray,
    [Out] System.Object^ Body2InterferedFaceArray,
    [Out] System.Object^ IntersectedBodyArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Body1

First body in interference detection (see **Remarks**)

Body2

Second body in interference detection (see **Remarks**)

CoincidentInterference

True to treat coincidence as interference, false to not

Body1InterferedFaceArray

Array of faces in the first body that interfered with the second body

Body2InterferedFaceArray

Array of faces in the second body that interfered with the first body

IntersectedBodyArray

Array of interfering bodies

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [Modeler::CheckInterferenceBetweenTwoBodies](#).

Example

[Check for Interference Between Two Bodies \(VBA\)](#)

Remarks

Before calling this method, transform Body1 and Body2 to their proper positions in the coordinate space of the top-level assembly:

1. Select the assembly components.
2. Use [IComponent2::Transform2](#) to get the transform of each component.
3. Call [IComponent2::GetBodies2](#) for each component.
4. Use [IBody2::ApplyTransform](#) on each body using the transforms in step 2.
5. Specify Body1 and Body2 with the bodies that have been transformed to their correct positions in the assembly.

See Also

[IModeler Interface](#)

[IModeler Members](#)

[InterferenceDetectionMgr Interface](#)

[IModeler::CheckInterference3 Method \(\)](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

Copies hole data from the source hole to the destination hole.

.NET Syntax

Visual Basic (Declaration)

```
Function CopyWizardHole( _  
    ByVal SourceHole As System.Object, _  
    ByVal DestinationHole As System.Object, _  
    ByVal RebuildOwner As System.Boolean _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SourceHole As System.Object  
Dim DestinationHole As System.Object  
Dim RebuildOwner As System.Boolean  
Dim value As System.Integer  
  
value = instance.CopyWizardHole(SourceHole, DestinationHole, RebuildOwner)
```

C#

```
System.int CopyWizardHole(  
    System.object SourceHole,  
    System.object DestinationHole,  
    System.bool RebuildOwner  
)
```

C++/CLI

```
System.int CopyWizardHole(  
& System.Object^ SourceHole,  
& System.Object^ DestinationHole,  
& System.bool RebuildOwner  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SourceHole

[Source hole](#)

DestinationHole

[Destination hole](#)

RebuildOwner

True rebuilds the model, false does not

Return Value

0 if the call generated an error; 1 if the call did not generate an error

Visual Basic for Applications (VBA) Syntax

See [Modeler::CopyWizardHole](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::ICopyWizardHole Method](#)

[IWizardHoleFeatureData2 Interface](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0



Creates an arc.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function CreateArc( _  
    ByVal Center As System.Object, _  
    ByVal Axis As System.Object, _  
    ByVal Radius As System.Double, _  
    ByVal StartPoint As System.Object, _  
    ByVal EndPoint As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Axis As System.Object  
Dim Radius As System.Double  
Dim StartPoint As System.Object  
Dim EndPoint As System.Object  
Dim value As System.Object  
  
value = instance.CreateArc(Center, Axis, Radius, StartPoint, EndPoint)
```

C#

```
System.object CreateArc(  
    System.object Center,  
    System.object Axis,  
    System.double Radius,  
    System.object StartPoint,  
    System.object EndPoint  
)
```

C++/CLI

```
System.Object^ CreateArc(  
& System.Object^ Center,  
& System.Object^ Axis,  
& System.double Radius,  
& System.Object^ StartPoint,  
& System.Object^ EndPoint  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array containing 3 doubles for location of the center of the arc (x,y,z)
Axis
Array containing 3 doubles (x,y,z)
Radius
Arc radius
StartPoint
Array of 3 doubles (x,y,z)
EndPoint
Array of 3 doubles (x,y,z)

Return Value

[Curve](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateArc](#).

Example

[Create Temporary Extruded Body \(C#\)](#)
[Create Temporary Extruded Body \(VB.NET\)](#)
[Create Temporary Extruded Body \(VBA\)](#)

Remarks

If your application uses [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#), then your application must also use [ICurve::CreateTrimmedCurve2](#) for the curves created by [IModeler::CreateArc](#) or [IModeler::ICreateArc](#) and [IModeler::CreateLine](#) or [IModeler::ICreateLine](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IBody2::AddProfileArc Method](#)
[IBody2::IAddProfileArc Method](#)
[IModeler::CreateEllipse Method](#)
[IModeler::ICreateEllipse Method](#)

SOLIDWORKS API Help

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CreateBodiesFromSheets Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodiesFromSheets Method (IModeler)

Obsolete. Superseded by [IModeler::CreateBodiesFromSheets2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodiesFromSheets( _  
    ByVal Sheets As System.Object, _  
    ByVal Options As System.Integer, _  
    ByRef Error As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Sheets As System.Object  
Dim Options As System.Integer  
Dim Error As System.Integer  
Dim value As System.Object  
  
value = instance.CreateBodiesFromSheets(Sheets, Options, Error)
```

C#

```
System.object CreateBodiesFromSheets(  
    System.object Sheets,  
    System.int Options,  
    out System.int Error  
)
```

C++/CLI

```
System.Object^ CreateBodiesFromSheets(  
& System.Object^ Sheets,  
& System.int Options,  
& [Out] System.int Error  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sheets

Options

Error

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodiesFromSheets](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateBodiesFromSheets2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodiesFromSheets2 Method (IModeler)

Sews sheets to make sheet (surface) or solid bodies.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodiesFromSheets2( _  
    ByVal Sheets As System.Object, _  
    ByVal Options As System.Integer, _  
    ByVal KnittingTolerance As System.Double, _  
    ByRef Error As System.Integer _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Sheets As System.Object  
Dim Options As System.Integer  
Dim KnittingTolerance As System.Double  
Dim Error As System.Integer  
Dim value As System.Object  
  
value = instance.CreateBodiesFromSheets2(Sheets, Options, KnittingTolerance, Error)
```

C#

```
System.object CreateBodiesFromSheets2(  
    System.object Sheets,  
    System.int Options,  
    System.double KnittingTolerance,  
    out System.int Error  
)
```

C++/CLI

```
System.Object^ CreateBodiesFromSheets2(  
& System.Object^ Sheets,  
& System.int Options,  
& System.double KnittingTolerance,  
& [Out] System.int Error  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Sheets

Array containing the [sheets](#)

Options

Type of body to create as defined by [swSheetSewingOption e](#)

KnittingTolerance

Knitting tolerance

Error

Error as defined by [swSheetSewingError e](#)

Return Value

Array of the sheet or solid [bodies](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodiesFromSheets2](#).

Remarks

It is safest to assume that the maximum number of bodies that can be returned by this method is the number of sheet input bodies.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateBrepBody3 Method](#)
[IModeler::ICreateWireBody Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateExtrudedBody Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

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CreateBodyFromBox Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodyFromBox Method (IModeler)

Obsolete. Superseded by [IModeler::CreateBodyFromBox3](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromBox( _  
    ByVal BoxDimArray As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim BoxDimArray As System.Object  
Dim value As System.Object  
  
value = instance.CreateBodyFromBox(BoxDimArray)
```

C#

```
System.object CreateBodyFromBox(  
    System.object BoxDimArray  
)
```

C++/CLI

```
System.Object^ CreateBodyFromBox(  
& System.Object^ BoxDimArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoxDimArray
Array of 9 doubles (see **Remarks**)

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodyFromBox](#).

Example

[Create Solid Bodies Using Geometry and Topology \(C#\)](#)
[Create Solid Bodies Using Geometry and Topology \(VB.NET\)](#)
[Create Solid Bodies Using Geometry and Topology \(VBA\)](#)

Remarks

The input parameter is the following array of doubles:

[*boxFaceCenter*[3], *boxAxis*[3], *boxWidth*, *boxLength*, *boxHeight*]

where:

<i>boxFaceCenter</i> [3]	XYZ location that represents the center of one of the box faces.
<i>boxAxis</i> [3]	XYZ direction. The box will be extruded along this vector from the <i>boxFaceCenter</i> location, a distance of <i>boxHeight</i> .
<i>boxWidth</i>	Box width. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the X-axis; otherwise, the orientation is not defined.
<i>boxLength</i>	Box length. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the Y axis; otherwise, the orientation is not defined.
<i>boxHeight</i>	Height to extrude along the <i>boxAxis</i> direction. If <i>boxHeight</i> is 0, a sheet body will be created of dimension <i>boxWidth</i> x <i>boxLength</i> and whose normal is defined by <i>boxAxis</i> .

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateExtrudedBody Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateBrepBody3 Method](#)
[IModeler::ICreateWireBody Method](#)

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CreateBodyFromBox3 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodyFromBox3 Method (IModeler)

Creates a temporary body from the specified box dimensions.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromBox3( _  
    ByVal BoxDimArray As System.Object _  
) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim BoxDimArray As System.Object
Dim value As Body2

value = instance.CreateBodyFromBox3(BoxDimArray)
```

C#

```
Body2 CreateBodyFromBox3(
    System.object BoxDimArray
)
```

C++/CLI

```
Body2^ CreateBodyFromBox3(
    & System.Object^ BoxDimArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

BoxDimArray
Array of 9 doubles (see **Remarks**)

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [IModeler::CreateBodyFromBox3](#).

Example

[Create Multibody Macro Feature \(VBA\)](#)
[Create Multibody Macro Feature \(VB.NET\)](#)
[Create Multibody Macro Feature \(C#\)](#)

Remarks

The input parameter is the following array of doubles:

$[\textit{boxFaceCenter}[3], \textit{boxAxis}[3], \textit{boxWidth}, \textit{boxLength}, \textit{boxHeight}]$

where:

<i>boxFaceCenter</i> [3]	XYZ location that represents the center of one of the box faces.
<i>boxAxis</i> [3]	XYZ direction. The box will be extruded along this vector from the <i>boxFaceCenter</i> location, a distance of <i>boxHeight</i> .
<i>boxWidth</i>	Box width. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the X-axis; the new body is rotated to the input <i>boxAxis</i> direction and translates it to the <i>boxFaceCenter</i> .
<i>boxLength</i>	Box length. If <i>boxAxis</i> is parallel to the Z axis (0,0,1), then this value represents the dimension that is parallel to the Y axis; the new body is rotated to the input <i>boxAxis</i> direction and translates it to the <i>boxFaceCenter</i> .
<i>boxHeight</i>	Height to extrude along the <i>boxAxis</i> direction. If <i>boxHeight</i> is 0, a sheet body and whose normal is defined by <i>boxAxis</i> .

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IFeature::ISetBody3 Method](#)
[IFeature::SetBody2 Method](#)

Availability

CreateBodyFromCone Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodyFromCone Method (IModeler)

Creates a temporary body from cone dimensions.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromCone( _  
    ByVal ConeDimArray As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim ConeDimArray As System.Object  
Dim value As System.Object  
  
value = instance.CreateBodyFromCone(ConeDimArray)
```

C#

```
System.object CreateBodyFromCone(  
    System.object ConeDimArray  
)
```

C++/CLI

```
System.Object^ CreateBodyFromCone(  
    & System.Object^ ConeDimArray  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConeDimArray
Array containing 9 doubles (see **Remarks**)

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodyFromCone](#).

Example

[Create Solid Bodies Using Geometry and Topology \(C#\)](#)
[Create Solid Bodies Using Geometry and Topology \(VB.NET\)](#)
[Create Solid Bodies Using Geometry and Topology \(VBA\)](#)

Remarks

The ConeArray argument is the following array of doubles:

[*coneFaceCenter*[3], *coneAxis*[3], *coneBaseRadius*, *coneTopRadius*, *coneHeight*]

where:

<i>coneFaceCenter</i> [3]	XYZ location that represents the center of one of the cone faces
<i>coneAxis</i> [3]	XYZ direction; the cone is extruded along this vector from the <i>coneFaceCenter</i> location, a distance of <i>coneHeight</i>

<i>coneBaseRadius</i>	Cone radius at the <i>coneFaceCenter</i> plane
<i>coneTopRadius</i>	Cone radius at <i>coneHeight</i> from the <i>coneFaceCenter</i> along the axis
<i>coneHeight</i>	Length to extrude along the <i>coneAxis</i> direction; if <i>coneHeight</i> is 0, then a sheet body is created of dimension <i>coneBaseRadius</i> and normal is defined by <i>coneAxis</i>

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateExtrudedBody Method](#)
[IModeler::CreateSweepBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateBrepBody3 Method](#)
[IModeler::ICreateWireBody Method](#)

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CreateBodyFromCyl Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodyFromCyl Method (IModeler)

Creates a temporary body from cylinder dimensions.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromCyl( _
    ByVal CylDimArray As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim CylDimArray As System.Object
Dim value As System.Object

value = instance.CreateBodyFromCyl(CylDimArray)
```

C#

```
System.object CreateBodyFromCyl(
    System.object CylDimArray
)
```

C++/CLI

```
System.Object^ CreateBodyFromCyl(
    & System.Object^ CylDimArray
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CylDimArray
Array containing 8 doubles (see **Remarks**)

Return Value

[Body](#).

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodyFromCyl](#).

Remarks

The `CylArray` argument is the following array of doubles:

[*cylFaceCenter*[3], *cylAxis*[3], *cylRadius*, *cylHeight*]

where:

<i>cylFaceCenter</i> [3]	XYZ location that represents the center of one of the cylinder faces
<i>cylAxis</i> [3]	XYZ direction; the cylinder is extruded along this vector from the <i>cylFaceCenter</i> location for a distance of <i>cylHeight</i>
<i>cylRadius</i>	Cylinder radius
<i>cylHeight</i>	Length to extrude along the <i>cylAxis</i> direction; if <i>cylHeight</i> is 0, then a sheet body is created of dimension <i>cylRadius</i> and whose normal is defined by <i>cylAxis</i>

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateExtrudedBody Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::ICreateBodiesFromSheets2 Method](#)
[IModeler::ICreateBodyFromBox2 Method](#)
[IModeler::ICreateBodyFromCone2 Method](#)
[IModeler::ICreateBodyFromCyl2 Method](#)
[IModeler::ICreateBodyFromFaces3 Method](#)
[IModeler::ICreateBrepBody3 Method](#)
[IModeler::ICreateWireBody Method](#)

SOLIDWORKS API Help

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CreateBodyFromFaces2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBodyFromFaces2 Method (IModeler)

Creates a temporary body from a list of faces.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBodyFromFaces2( _  
    ByVal NumOfFaces As System.Integer, _  
    ByVal Faces As System.Object, _  
    ByVal ActionType As System.Integer, _  
    ByVal DoLocalCheck As System.Boolean, _  
    ByRef LocalCheckResult As System.Boolean _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim NumOfFaces As System.Integer  
Dim Faces As System.Object  
Dim ActionType As System.Integer  
Dim DoLocalCheck As System.Boolean  
Dim LocalCheckResult As System.Boolean  
Dim value As System.Object  
  
value = instance.CreateBodyFromFaces2(NumOfFaces, Faces, ActionType, DoLocalCheck,  
LocalCheckResult)
```

C#

```
System.object CreateBodyFromFaces2(
    System.int NumOfFaces,
    System.object Faces,
    System.int ActionType,
    System.bool DoLocalCheck,
    out System.bool LocalCheckResult
)
```

C++/CLI

```
System.Object^ CreateBodyFromFaces2(
    & System.int NumOfFaces,
    & System.Object^ Faces,
    & System.int ActionType,
    & System.bool DoLocalCheck,
    [Out] System.bool LocalCheckResult
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NumOfFaces

Number of faces

Faces

Array of the [faces](#)

ActionType

Type of action as defined in [swCreateFacesBodyAction_e](#)

DoLocalCheck

True to perform local checking on the resulting body, false to not

LocalCheckResult

If DoLocalCheck is True, then True if body is okay, false if not

Return Value

[Body](#) or NULL if the operation fails

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBodyFromFaces2](#).

Remarks

This method is useful when you attempt to copy a body with changes to that body. All of the input faces must be from the same body.

If you call this method to remove a hole, then the value that you specify for ActionType indicates to the modeler how to handle filling the hole. The result must be a solid.

This method is the opposite of [IBody2::DeleteFaces3](#) or [IBody2::IDeleteFaces3](#), which allows you to specify faces to remove.

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IModeler::CreateBodiesFromSheets2 Method](#)

[IModeler::CreateBodyFromBox Method](#)

[IModeler::CreateBodyFromCone Method](#)

[IModeler::CreateBodyFromCyl Method](#)

[IModeler::CreateBrepBody3 Method](#)

[IModeler::CreateExtrudedBody Method](#)

[IModeler::CreateSweptBody Method](#)

[IModeler::CreateWireBody Method](#)

[IModeler::ICreateBodiesFromSheets2 Method](#)

[IModeler::ICreateBodyFromBox2 Method](#)

[IModeler::ICreateBodyFromCone2 Method](#)

[IModeler::ICreateBodyFromCyl2 Method](#)

[IModeler::ICreateBodyFromFaces3 Method](#)

[IModeler::ICreateBrepBody3 Method](#)

[IModeler::ICreateWireBody Method](#)

SOLIDWORKS API Help

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CreateBrepBody Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBrepBody Method (IModeler)

Obsolete. Superseded by [IModeler::CreateBrepBody3](#).

Visual Basic (Declaration)

```
Function CreateBrepBody( _
    ByVal Type As System.Integer, _
    ByVal NTopologies As System.Integer, _
    ByVal Topologies As System.Object, _
    ByVal EdgeToleranceArray As System.Object, _
    ByVal VertexToleranceArray As System.Object, _
    ByVal PointArray As System.Object, _
    ByVal CurveArray As System.Object, _
    ByVal SurfaceArray As System.Object, _
    ByVal NRelations As System.Integer, _
    ByVal Parents As System.Object, _
    ByVal Children As System.Object, _
    ByVal Senses As System.Object _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim Type As System.Integer
Dim NTopologies As System.Integer
Dim Topologies As System.Object
Dim EdgeToleranceArray As System.Object
Dim VertexToleranceArray As System.Object
Dim PointArray As System.Object
Dim CurveArray As System.Object
Dim SurfaceArray As System.Object
Dim NRelations As System.Integer
Dim Parents As System.Object
Dim Children As System.Object
Dim Senses As System.Object
Dim value As System.Object

value = instance.CreateBrepBody(Type, NTopologies, Topologies, EdgeToleranceArray,
VertexToleranceArray, PointArray, CurveArray, SurfaceArray, NRelations, Parents, Children, Senses)
```

C#

```
System.object CreateBrepBody(
    System.int Type,
    System.int NTopologies,
    System.object Topologies,
    System.object EdgeToleranceArray,
    System.object VertexToleranceArray,
    System.object PointArray,
    System.object CurveArray,
    System.object SurfaceArray,
    System.int NRelations,
    System.object Parents,
    System.object Children,
    System.object Senses
)
```

C++/CLI

```
System.Object^ CreateBrepBody(
    & System.int Type,
    & System.int NTopologies,
    & System.Object^ Topologies,
    & System.Object^ EdgeToleranceArray,
    & System.Object^ VertexToleranceArray,
    & System.Object^ PointArray,
    & System.Object^ CurveArray,
    & System.Object^ SurfaceArray,
    & System.int NRelations,
    & System.Object^ Parents,
```

```
& System.Object^ Children,  
& System.Object^ Senses  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

NTopologies

Topologies

EdgeToleranceArray

VertexToleranceArray

PointArray

CurveArray

SurfaceArray

NRelations

Parents

Children

Senses

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBrepBody](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateBrepBody3 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBrepBody3 Method (IModeler)

Creates a temporary body from BREP (boundary representation) data.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBrepBody3(  
    ByVal Type As System.Integer, _  
    ByVal NTopologies As System.Integer, _  
    ByVal Topologies As System.Object, _  
    ByVal EdgeToleranceArray As System.Object, _  
    ByVal VertexToleranceArray As System.Object, _  
    ByVal PointArray As System.Object, _  
    ByVal CurveArray1 As System.Object, _  
    ByVal CurveSurfaceArray1 As System.Object, _  
    ByVal CurveArray2 As System.Object, _  
    ByVal CurveSurfaceArray2 As System.Object, _  
    ByVal SurfaceArray As System.Object, _  
    ByVal NRelations As System.Integer, _  
    ByVal Parents As System.Object, _  
    ByVal Children As System.Object, _  
    ByVal Senses As System.Object, _  
    ByVal Option As System.Integer _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim Type As System.Integer  
Dim NTopologies As System.Integer
```



```

Dim Topologies As System.Object
Dim EdgeToleranceArray As System.Object
Dim VertexToleranceArray As System.Object
Dim PointArray As System.Object
Dim CurveArray1 As System.Object
Dim CurveSurfaceArray1 As System.Object
Dim CurveArray2 As System.Object
Dim CurveSurfaceArray2 As System.Object
Dim SurfaceArray As System.Object
Dim NRelations As System.Integer
Dim Parents As System.Object
Dim Children As System.Object
Dim Senses As System.Object
Dim Option As System.Integer
Dim value As System.Object

```

```

value = instance.CreateBrepBody3(Type, NTopologies, Topologies, EdgeToleranceArray,
VertexToleranceArray, PointArray, CurveArray1, CurveSurfaceArray1, CurveArray2,
CurveSurfaceArray2, SurfaceArray, NRelations, Parents, Children, Senses, Option)

```

C#

```

System.object CreateBrepBody3(
    System.int Type,
    System.int NTopologies,
    System.object Topologies,
    System.object EdgeToleranceArray,
    System.object VertexToleranceArray,
    System.object PointArray,
    System.object CurveArray1,
    System.object CurveSurfaceArray1,
    System.object CurveArray2,
    System.object CurveSurfaceArray2,
    System.object SurfaceArray,
    System.int NRelations,
    System.object Parents,
    System.object Children,
    System.object Senses,
    System.int Option
)

```

C++/CLI

```

System.Object^ CreateBrepBody3(
    & System.int Type,
    & System.int NTopologies,
    & System.Object^ Topologies,
    & System.Object^ EdgeToleranceArray,
    & System.Object^ VertexToleranceArray,
    & System.Object^ PointArray,
    & System.Object^ CurveArray1,
    & System.Object^ CurveSurfaceArray1,
    & System.Object^ CurveArray2,
    & System.Object^ CurveSurfaceArray2,
    & System.Object^ SurfaceArray,
    & System.int NRelations,
    & System.Object^ Parents,
    & System.Object^ Children,
    & System.Object^ Senses,
    & System.int Option
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type

Type of body to create as defined in [swTopology_e](#)

NTopologies

Number of topological entities in the topologies argument

Topologies

Array of topologies (see [swTopoEntity_e](#)), one for each topological entity

EdgeToleranceArray

Array of tolerances for edges

VertexToleranceArray
 Array of tolerances for vertices
PointArray
 Array of coordinates of vertices (geometry for vertices)
CurveArray1
 Array of [curves](#) (geometry for edges; 3D curve) and coedges (geometry for coedges; 2D curve)
CurveSurfaceArray1
 Array of [surfaces](#) that lie on CurveArray1 2D curve
CurveArray2
 Array of [curves](#) (geometry for edges; 3D curve) and array of coedges (geometry for coedges; 2D curve)
CurveSurfaceArray2
 Array of [surfaces](#) that on lie on CurveArray2 2D curve
SurfaceArray
 Array of [surfaces](#) (geometry for faces)
NRelations
 Number of 1-to-1 relationships between topological entities
Parents
 Array of parents, one in each relationship
Children
 Array of parents, one in each relationship
Senses
 Array of senses in which child is used by parent in the relationship
Option

- 0 = Default

- 1 = Repair and simplify body
- 2 = Simplify body

Return Value

[Body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBrepBody3](#).

Remarks

The CurveArray1 and CurveArray2 must be paired with CurveSurfaceArray1 and CurveSurfaceArray2, respectively. Order is not important. The 2D curve has to be created in the direction of the loop.

If non-negative values are packed into the EdgeToleranceArray and VertexToleranceArray arrays, then tolerances are applied to the corresponding edges or vertices. These arrays should be the same size as CurveArray1 and PointArray, respectively. Otherwise, a default value of 1.0e-8 (modeler precision) is used.

NOTE: [IModeler::SetInitKnitGapWidth](#) does not affect this method.

Useful methods for creating geometry for the topological entities are:

- [IBody2::CreatePlanarSurface](#) or [IBody2::ICreatePlanarSurface](#)
- [IBody2::AddProfileArc](#) or [IBody2::IAddProfileArc](#)
- [IBody2::AddProfileLine](#) or [IBody2::IAddProfileLine](#)
- [IBody2::CreateRevolutionSurface](#) or [IBody2::ICreateRevolutionSurface](#)

For example, to create a cone, find topological relationships and form relevant arrays. A complete solid cone consists of 8 topological entities:

- 1 shell
- 2 faces
- 3 loops
- 1 edge
- 1 vertex

There are 8 relations:

- the shell is the parent of 2 faces - 2
- the planar end face has 1 loop - 1
- the conical face has 2 loops - 2
- two loops are each the parent of 1 edge - 2
- one loop is the parent of 1 vertex 1

The topologies array:

Index	Value
0	swTopoShell
1	swTopoFace
2	swTopoFace

3	swTopoLoop
4	swTopoEdge
5	swTopoLoop
6	swTopoLoop
7	swTopoVertex

The set of arrays:

index	parents	children	senses	relation
0	0	1	0	shell to face
1	0	2	0	shell to face
2	1	3	0	face to loop
3	3	4	-1	loop to edge
4	2	5	0	face to loop
5	2	6	0	face to loop
6	5	4	1	loop to edge
7	6	7	0	loop to vertex

Values in the parents and children arrays correspond to the indices of the topology array.

Each face or edge created by `IModeler::CreateBrepBody3` or [IModeler::ICreateBrepBody3](#) has an associated integer ID that is the same as the index to the input topologies. Use [IFace2::GetFaceId](#) or [IEdge::GetId](#) to get the associated integer ID.

Every shell should be a closed shell. Sheet bodies should have additional back faces to form a closed shell. When creating a sheet body, these extra back faces are retained in the result and should be removed using [IModeler::DeleteFacesFromSheetBody](#) or [IModeler::!DeleteFacesFromSheetBody](#).

 See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::!CreateBodiesFromSheets2 Method](#)
[IModeler::!CreateBodyFromBox2 Method](#)
[IModeler::!CreateBodyFromCone2 Method](#)
[IModeler::!CreateBodyFromCyl2 Method](#)
[IModeler::!CreateBodyFromFaces3 Method](#)
[IModeler::!CreateWireBody Method](#)

 Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateBsplineCurve Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBsplineCurve Method (IModeler)

Creates a b-spline curve.

 .NET Syntax

Visual Basic (Declaration)

```
Function CreateBsplineCurve( _  
    ByVal Props As System.Object, _
```

```
ByVal Knots As System.Object, _  
ByVal CtrlPtCoords As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Props As System.Object  
Dim Knots As System.Object  
Dim CtrlPtCoords As System.Object  
Dim value As System.Object  
  
value = instance.CreateBsplineCurve(Props, Knots, CtrlPtCoords)
```

C#

```
System.object CreateBsplineCurve(  
    System.object Props,  
    System.object Knots,  
    System.object CtrlPtCoords  
)
```

C++/CLI

```
System.Object^ CreateBsplineCurve(  
& System.Object^ Props,  
& System.Object^ Knots,  
& System.Object^ CtrlPtCoords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Props
Array containing 4 integers packed into 2 double elements (see **Remarks**)
Knots
Array of numKnots doubles (see **Remarks**)
CtrlPtCoords
Array of NumCtrlPtCoord doubles (see **Remarks**)

Return Value

B-spline [curve](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBsplineCurve](#).

Remarks

The Props argument contains the following values:

- DimensionControlPoints
- Order
- NumCtrlPoints
- Periodicity

Length of this array...	Given by...
Knots	numKnots = NumCtrlPoints + Order
CtrlPtCoords	NumCtrlPtCoord = NumCtrlPoints + DimensionControlPoints

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::ICreateBsplineCurve Method](#)
[IBody2::AddProfileBspline Method](#)
[IBody2::IAddProfileBspline Method](#)

CreateBsplinesurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateBsplinesurface Method (IModeler)

Creates a b-spline surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateBsplinesurface( _  
    ByVal Props As System.Object, _  
    ByVal UKnots As System.Object, _  
    ByVal VKnots As System.Object, _  
    ByVal CtrlPtCoords As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Props As System.Object  
Dim UKnots As System.Object  
Dim VKnots As System.Object  
Dim CtrlPtCoords As System.Object  
Dim value As System.Object  
  
value = instance.CreateBsplinesurface(Props, UKnots, VKnots, CtrlPtCoords)
```

C#

```
System.object CreateBsplinesurface(  
    System.object Props,  
    System.object UKnots,  
    System.object VKnots,  
    System.object CtrlPtCoords  
)
```

C++/CLI

```
System.Object^ CreateBsplinesurface(  
& System.Object^ Props,  
& System.Object^ UKnots,  
& System.Object^ VKnots,  
& System.Object^ CtrlPtCoords  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Props
Array containing 8 integers packed as 4 double elements (see **Remarks**)
UKnots
Array of numUKnots (see **Remarks**)
VKnots
Array of numVKnots (see **Remarks**)
CtrlPtCoords
Array of numCtrlPtCoord (see **Remarks**)

Return Value

B-spline [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateBsplinesurface](#).

Remarks

The Props array contains the following elements. These are integers packed into a double array in the Dispatch version.

- Uorder, Vorder

- numV_CtrlPoints, numU_CtrlPoints
- Uperiodicity, Vperiodicity
- DimensionControlPoints, UNUSED (set =0)

Number of elements in this array...	Given by...
UKnots	numUKnots = numU_CtrlPoints + Uorder
VKnots	numVKnots = numV_CtrlPoints + Vorder
CtrlPtCoords	numCtrlPtCoord =(NumV_CtrlPoints * NumU_CtrlPoints * DimensionControlPoints)

NOTE: The order of the UV control points are reversed if you used [ISurface::GetBSurfParams3](#) or [ISurface::IGetBSurfParams](#) to get the data. You do not need to reverse the UV control points; instead, you can switch the UV knots and related parameters.

For periodics, the first knot must have multiplicity = 1 and all excess multiplicity must be imposed on the last knot. Therefore, a valid periodic knv vector would be { 0, 1, 2, 3, 3, 3 } for a cubic curve. The control point on the seam of the closed curve cannot be replicated at both ends; it should occur only at the beginning.

To convert from a clamped periodic curve (numKnots = numCtrlPts + Order, ctrlPts replicated, knot multiplicity = order at each end) to a SOLIDWORKS periodic curve, remove all but one of the knots at the beginning of the vector and remove one at the end. Also remove the last control point to avoid the point replication.

 See Also

- [IModeler Interface](#)
- [IModeler Members](#)
- [IModeler::ICreateBsplineSurface Method](#)
- [IBody2::CreateBsplineSurface Method](#)
- [IBody2::ICreateBsplineSurface Method](#)
- [IModeler::CreateConicalSurface2 Method](#)
- [IModeler::CreateCoonsBSurface Method](#)
- [IModeler::CreateCylindricalSurface2 Method](#)
- [IModeler::CreateExtrusionSurface Method](#)
- [IModeler::CreateLoftSurface Method](#)
- [IModeler::CreateOffsetSurface Method](#)
- [IModeler::CreatePlanarSurface2 Method](#)
- [IModeler::CreateRuledSurface Method](#)
- [IModeler::CreateSphericalSurface2 Method](#)
- [IModeler::CreateSweptSurface Method](#)
- [IModeler::CreateToroidalSurface Method](#)
- [IModeler::ICreateConicalSurface2 Method](#)
- [IModeler::ICreateCylindricalSurface2 Method](#)
- [IModeler::ICreateExtrusionSurface Method](#)
- [IModeler::ICreateLoftSurface Method](#)
- [IModeler::ICreateOffsetSurface Method](#)
- [IModeler::ICreatePlanarSurface2 Method](#)
- [IModeler::ICreateRuledSurface Method](#)
- [IModeler::ICreateSphericalSurface2 Method](#)
- [IModeler::ICreateSweptSurface Method](#)
- [IModeler::ICreateToroidalSurface Method](#)
- [IModeler::CreateBsplineCurve Method](#)
- [IModeler::CreateBsplineCurve Method](#)

CreateConicalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateConicalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::CreateConicalSurface2](#).

 .NET Syntax

Visual Basic (Declaration)

```
Function CreateConicalSurface( _  
    ByVal Center As System.Object, _  
    ByVal Direction As System.Object, _  
    ByVal Radius As System.Double, _  
    ByVal SemiAngle As System.Double _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Direction As System.Object
```



```
Dim Radius As System.Double
Dim SemiAngle As System.Double
Dim value As System.Object

value = instance.CreateConicalSurface(Center, Direction, Radius, SemiAngle)
```

C#

```
System.Object CreateConicalSurface(
    System.Object Center,
    System.Object Direction,
    System.Double Radius,
    System.Double SemiAngle
)
```

C++/CLI

```
System.Object^ CreateConicalSurface(
    & System.Object^ Center,
    & System.Object^ Direction,
    & System.Double Radius,
    & System.Double SemiAngle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Direction

Radius

SemiAngle

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateConicalSurface](#).

See Also

[IModeler Interface](#)

[IModeler Members](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

CreateConicalSurface2 Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateConicalSurface2 Method (IModeler)

Creates an untrimmed conical surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateConicalSurface2( _
    ByVal Center As System.Object, _
    ByVal Direction As System.Object, _
    ByVal RefDirection As System.Object, _
    ByVal Radius As System.Double, _
    ByVal SemiAngle As System.Double _
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModeler
Dim Center As System.Object
Dim Direction As System.Object
Dim RefDirection As System.Object
```



```
Dim Radius As System.Double
Dim SemiAngle As System.Double
Dim value As System.Object
```

```
value = instance.CreateConicalSurface2(Center, Direction, RefDirection, Radius, SemiAngle)
```

C#

```
System.Object CreateConicalSurface2(
    System.Object Center,
    System.Object Direction,
    System.Object RefDirection,
    System.Double Radius,
    System.Double SemiAngle
)
```

C++/CLI

```
System.Object^ CreateConicalSurface2(
    & System.Object^ Center,
    & System.Object^ Direction,
    & System.Object^ RefDirection,
    & System.Double Radius,
    & System.Double SemiAngle
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center
Array containing 3 doubles (see **Remarks**)
Direction
Array containing 3 doubles (see **Remarks**)
RefDirection
Array containing 3 doubles (see **Remarks**)
Radius
See **Remarks**
SemiAngle
See **Remarks**

Return Value

[Surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateConicalSurface2](#).

Remarks

Input arguments:

Argument	Description
Center	XYZ location which represents the center of the bottom
Direction	XYZ direction of the axis of the conical surface
Radius	Radius at the center
SemiAngle	Half angle of the cone in radians

You can trim the resulting surface to generate a sheet body (for example, using [ISurface::CreateTrimmedSheet](#)).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)

[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2001Plus FCS, Revision Number 10.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCoonsBSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateCoonsBSurface Method (IModeler)

Creates a coons b-surface using the four specified boundary curves.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCoonsBSurface( _  
    ByVal UCurve0 As Curve, _  
    ByVal UCurve1 As Curve, _  
    ByVal VCurve0 As Curve, _  
    ByVal VCurve1 As Curve, _  
    ByVal IgnoreClosureReq As System.Boolean _  
) As Surface
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim UCurve0 As Curve  
Dim UCurve1 As Curve  
Dim VCurve0 As Curve  
Dim VCurve1 As Curve  
Dim IgnoreClosureReq As System.Boolean  
Dim value As Surface  
  
value = instance.CreateCoonsBSurface(UCurve0, UCurve1, VCurve0, VCurve1, IgnoreClosureReq)
```

C#

```
Surface CreateCoonsBSurface(  
    Curve UCurve0,  
    Curve UCurve1,  
    Curve VCurve0,  
    Curve VCurve1,  
    System.bool IgnoreClosureReq  
)
```

C++/CLI

```
Surface^ CreateCoonsBSurface(  
& Curve^ UCurve0,  
& Curve^ UCurve1,  
& Curve^ VCurve0,  
& Curve^ VCurve1,  
& System.bool IgnoreClosureReq  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UCurve0
First curve of u direction
UCurve1

Second curve of u direction

VCurve0

First curve of v direction

VCurve1

Second curve of v direction

IgnoreClosureReq

True to ignore closure request, false to not

Return Value

[B-surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateCoonsBSurface](#).

Remarks

By default, IgnoreClosureReq is not ignored.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSheetFromSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplineSurface Method](#)
[IModeler::CreateBsplineSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCylindricalSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateCylindricalSurface Method (IModeler)

Obsolete. Superseded by [IModeler::CreateCylindricalSurface2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCylindricalSurface( _  
    ByVal Center As System.Object, _  
    ByVal Direction As System.Object, _  
    ByVal Radius As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Direction As System.Object  
Dim Radius As System.Double
```

```
Dim value As System.Object
```

```
value = instance.CreateCylindricalSurface(Center, Direction, Radius)
```

C#

```
System.Object CreateCylindricalSurface(  
    System.Object Center,  
    System.Object Direction,  
    System.Double Radius  
)
```

C++/CLI

```
System.Object^ CreateCylindricalSurface(  
& System.Object^ Center,  
& System.Object^ Direction,  
& System.Double Radius  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Direction

Radius

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateCylindricalSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateCylindricalSurface2 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateCylindricalSurface2 Method (IModeler)

Creates an untrimmed cylindrical surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateCylindricalSurface2( _  
    ByVal Center As System.Object, _  
    ByVal Direction As System.Object, _  
    ByVal RefDirection As System.Object, _  
    ByVal Radius As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim Direction As System.Object  
Dim RefDirection As System.Object  
Dim Radius As System.Double  
Dim value As System.Object  
  
value = instance.CreateCylindricalSurface2(Center, Direction, RefDirection, Radius)
```

C#

```
System.object CreateCylindricalSurface2(
    System.object Center,
    System.object Direction,
    System.object RefDirection,
    System.double Radius
)
```

C++/CLI

```
System.Object^ CreateCylindricalSurface2(
    & System.Object^ Center,
    & System.Object^ Direction,
    & System.Object^ RefDirection,
    & System.double Radius
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Array containing 3 doubles for XYZ location of the center of the bottom

Direction

Array containing 3 doubles for XYZ direction of the axis of the cylindrical surface

RefDirection

Array containing 3 doubles for XYZ direction of the axis of the cylindrical surface

Radius

Radius at the center

Return Value

[Surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateCylindricalSurface2](#).

Example

[Create Spherical Surface, Trimmed Sheet, and Feature from Body \(VBA\)](#)

Remarks

You can trim the resulting surface to generate a sheet body, for example, using [ISurface::CreateTrimmedSheet](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSheetFromSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

CreateEllipse Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateEllipse Method (IModeler)

Creates a temporary elliptical curve.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateEllipse( _  
    ByVal Center As System.Object, _  
    ByVal MajorRadius As System.Double, _  
    ByVal MinorRadius As System.Double, _  
    ByVal MajorAxis As System.Object, _  
    ByVal MinorAxis As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim Center As System.Object  
Dim MajorRadius As System.Double  
Dim MinorRadius As System.Double  
Dim MajorAxis As System.Object  
Dim MinorAxis As System.Object  
Dim value As System.Object  
  
value = instance.CreateEllipse(Center, MajorRadius, MinorRadius, MajorAxis, MinorAxis)
```

C#

```
System.object CreateEllipse(  
    System.object Center,  
    System.double MajorRadius,  
    System.double MinorRadius,  
    System.object MajorAxis,  
    System.object MinorAxis  
)
```

C++/CLI

```
System.Object^ CreateEllipse(  
& System.Object^ Center,  
& System.double MajorRadius,  
& System.double MinorRadius,  
& System.Object^ MajorAxis,  
& System.Object^ MinorAxis  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Center

Array of 3 doubles describing the center of the ellipse

MajorRadius

Major radius of ellipse

MinorRadius

Minor radius of ellipse

MajorAxis

Array of 3 doubles describing the major axis of the ellipse

MinorAxis

Array of 3 doubles describing the minor axis of the ellipse

Return Value

[Curve](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateEllipse](#).

Example

[Create Temporary Elliptical Extrusion \(VBA\)](#)
[Create Temporary Elliptical Extrusion \(VB.NET\)](#)
[Create Temporary Elliptical Extrusion \(C#\)](#)

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateArc Method](#)
[IModeler::ICreateArc Method](#)
[IModeler::CreateLine Method](#)
[IModeler::ICreateLine Method](#)
[IModeler::ICreateEllipse Method](#)
[IModeler::CreateExtrudedBody Method](#)

Availability

SOLIDWORKS 2012 SP01, Revision Number 20.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateExtrudedBody Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateExtrudedBody Method (IModeler)

Creates a temporary extruded body.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateExtrudedBody( _  
    ByVal SheetBody As Body2, _  
    ByVal Direction As MathVector, _  
    ByVal Length As System.Double _  
    ) As Body2
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SheetBody As Body2  
Dim Direction As MathVector  
Dim Length As System.Double  
Dim value As Body2  
  
value = instance.CreateExtrudedBody(SheetBody, Direction, Length)
```

C#

```
Body2 CreateExtrudedBody(  
    Body2 SheetBody,  
    MathVector Direction,  
    System.double Length  
)
```

C++/CLI

```
Body2^ CreateExtrudedBody(  
& Body2^ SheetBody,  
& MathVector^ Direction,  
& System.double Length  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SheetBody

Sheet body that defines the profile for the temporary extruded body (see **Remarks**)

Direction

Direction of the axis for the temporary extruded body

Length

Length of the temporary extruded body

Return Value

Temporary extruded [body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateExtrudedBody](#).

Example

[Create Temporary Elliptical Extrusion \(C#\)](#)
[Create Temporary Elliptical Extrusion \(VB.NET\)](#)
[Create Temporary Elliptical Extrusion \(VBA\)](#)
[Create Temporary Extruded Body \(C#\)](#)
[Create Temporary Extruded Body \(VB.NET\)](#)
[Create Temporary Extruded Body \(VBA\)](#)

Remarks

Before calling this method:

1. Call [IModeler::CreateArc](#), [IModeler::CreateArc](#), [IModeler::CreateEllipse](#), [IModeler::CreateEllipse](#), [IModeler::CreateLine](#), or [IModeler::CreateLine](#) to create curves that define the profile of the body that you want to extrude.
2. Call [ICurve::CreateTrimmedCurve2](#) to trim the curves of arcs and lines only.
3. Call [ISurface::CreateTrimmedSheet4](#) or [ISurface::CreateTrimmedSheet4](#) to create the profile body, passing in either trimmed arcs and lines or untrimmed elliptical curves.
4. Set SheetBody to the profile body.

After calling this method, call [IBody2::Display3](#) to display the extrusion.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox Method](#)
[IModeler::CreateBodyFromCone Method](#)
[IModeler::CreateBodyFromCyl Method](#)
[IModeler::CreateBodyFromFaces2 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateSweptBody Method](#)
[IModeler::CreateWireBody Method](#)
[IModeler::CreateBodiesFromSheets2 Method](#)
[IModeler::CreateBodyFromBox2 Method](#)
[IModeler::CreateBodyFromCone2 Method](#)
[IModeler::CreateBodyFromCyl2 Method](#)
[IModeler::CreateBodyFromFaces3 Method](#)
[IModeler::CreateBrepBody3 Method](#)
[IModeler::CreateWireBody Method](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateExtrusionSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateExtrusionSurface Method (IModeler)

Creates an extruded surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateExtrusionSurface( _  
    ByVal ProfileCurve As System.Object, _  
    ByVal AxisDirection As System.Object _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModeler  
Dim ProfileCurve As System.Object  
Dim AxisDirection As System.Object
```



```
Dim value As System.Object
```

```
value = instance.CreateExtrusionSurface(ProfileCurve, AxisDirection)
```

C#

```
System.Object CreateExtrusionSurface(  
    System.Object ProfileCurve,  
    System.Object AxisDirection  
)
```

C++/CLI

```
System.Object^ CreateExtrusionSurface(  
& System.Object^ ProfileCurve,  
& System.Object^ AxisDirection  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ProfileCurve
Profile [curve](#)
AxisDirection
Array of 3 doubles (x,y,z)

Return Value

Extruded [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateExtrusionSurface](#).

Remarks

You can use this method with:

- A set of related functions that construct a body from trimmed surfaces. The profile curve is extruded along the direction vector of axis direction, the new surface being the envelope of the curve. The profile curve must be of type line, circle, or b-spline curve.
- Trimming curve creation routines (for example, [ISurface::AddTrimmingLoop2](#) or [ISurface::IAddTrimmingLoop2](#)) to construct a trimmed tabulated cylinder.

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IBody2::CreateExtrusionSurface Method](#)
[IBody2::ICreateExtrusionSurface Method](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

CreateLine Method (IModeler)

[See Also](#) [Example](#)



Visual Basic Syntax

Visual Basic (Declaration)

```
Function CreateLine( _  
    ByVal RootPoint As System.Object, _  
    ByVal Direction As System.Object _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim RootPoint As System.Object  
Dim Direction As System.Object  
Dim value As System.Object  
  
value = instance.CreateLine(RootPoint, Direction)
```

C#

```
System.object CreateLine(  
    System.object RootPoint,  
    System.object Direction  
)
```

C++/CLI

```
System.Object^ CreateLine(  
& System.Object^ RootPoint,  
& System.Object^ Direction  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

RootPoint

Array containing 3 doubles (x, y, z) for the start point of the line

Direction

Array containing 3 doubles (x, y, z) for the end point of the line

Return Value

Newly created [line](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateLine](#).

Example

[Create Temporary Extruded Body \(C#\)](#)
[Create Temporary Extruded Body \(VB.NET\)](#)
[Create Temporary Extruded Body \(VBA\)](#)

Remarks

If your application uses [ISurface::CreateTrimmedSheet4](#) or [ISurface::ICreateTrimmedSheet4](#), then your application must also use [ICurve::CreateTrimmedCurve2](#) for the curves created by [IModeler::CreateArc](#) or [IModeler::ICreateArc](#) and [IModeler::CreateLine](#) or [IModeler::ICreateLine](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateEllipse Method](#)
[IModeler::ICreateEllipse Method](#)

Obsolete. Superseded by [IModeler::CreateLoftBody2](#).

.NET Syntax

Visual Basic (Declaration)

```
Function CreateLoftBody( _
    ByVal PModDoc As ModelDoc2, _
    ByVal IsBlendClosed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal IsSolidBody As System.Boolean, _
    ByVal TessTolFactor As System.Double, _
    ByVal StartMatchingType As System.Short, _
    ByVal EndMatchingType As System.Short _
) As Body2
```

Visual Basic (Usage)

```
Dim instance As IModeler
Dim PModDoc As ModelDoc2
Dim IsBlendClosed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim IsSolidBody As System.Boolean
Dim TessTolFactor As System.Double
Dim StartMatchingType As System.Short
Dim EndMatchingType As System.Short
Dim value As Body2
```

```
value = instance.CreateLoftBody(PModDoc, IsBlendClosed, KeepTangency, ForceNonRational,
IsSolidBody, TessTolFactor, StartMatchingType, EndMatchingType)
```

C#

```
Body2 CreateLoftBody(
    ModelDoc2 PModDoc,
    System.bool IsBlendClosed,
    System.bool KeepTangency,
    System.bool ForceNonRational,
    System.bool IsSolidBody,
    System.double TessTolFactor,
    System.short StartMatchingType,
    System.short EndMatchingType
)
```

C++/CLI

```
Body2^ CreateLoftBody(
    & ModelDoc2^ PModDoc,
    & System.bool IsBlendClosed,
    & System.bool KeepTangency,
    & System.bool ForceNonRational,
    & System.bool IsSolidBody,
    & System.double TessTolFactor,
    & System.short StartMatchingType,
    & System.short EndMatchingType
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PModDoc

[Model document](#) in which to create the lofted body

IsBlendClosed

True for a closed, false for an open loft; if true and you selected less than three profiles; any selected guide curves must be closed curves

KeepTangency

If the section curves are tangent, then you have the option to specify whether the resulting faces are also tangent; specify true to maintain the tangency as seen in the section curves, false to not

NOTE: When generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface shapes if the section curves exhibit these characteristics.

ForceNonRational

True to force the resulting surface to be non-rational, false to not

IsSolidBody

True to return a solid body, false to return a surface body

TessTolFactor

Factor to control the number of intermediate sections used for loft with centerline; default value is 1.0; the greater the variable, the more intermediate sections created

StartMatchingType

Tangency type at the start profile (see **Remarks**)

EndMatchingType

Tangency type as the end profile (see **Remarks**)

Return Value

Newly created [loft](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateLoftBody](#).

Remarks

Selection of guide curves and centerline is optional. However, you must select the profiles in an order consistent with the desired direction of the loft. Because a solid is being created, the section profiles must be closed.

It is best to use guide curves, especially when you select profiles in the FeatureManager design tree.

You can use any number of profiles. However, if you select only one profile, then any selected guide curves must be closed curves.

Use [IModelDocExtension::SelectByID2](#) to select the profiles and guide curves. Set the mark for:

- profile selections to 1
- any guide curve selection to 2
- centerline selection to 4
- start tangency vector selection to 8
- start tangency faces selection to 16 (not available)
- end tangency vector selection to 32
- end tangency faces selection to 64 (not available)

Linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

The tangency types can be:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to the selected vector
- 3 = tangent to all adjacent faces sharing an edge with the start profile

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IFeatureManager::InsertCutBlend Method](#)

[IFeatureManager::InsertProtrusionBlend Method](#)

[IModelDoc2::InsertLoftRefSurface2 Method](#)

[IModeler::CreateLoftSurface Method](#)

[IModeler::ICreateLoftSurface Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLoftBody2 Method (IModeler)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateLoftBody2 Method (IModeler)

Creates a loft body using the specified profiles, guide curves, and centerline.

I.NET Syntax

```

Function CreateLoftBody2( _
    ByVal PModDoc As ModelDoc2, _
    ByVal Profile As System.Object, _
    ByVal GuideCurve As System.Object, _
    ByVal Centerline As System.Object, _
    ByVal IsThinBody As System.Boolean, _
    ByVal ThinType As System.Integer, _
    ByVal Thickness1 As System.Double, _
    ByVal Thickness2 As System.Double, _
    ByVal FeatureScope As System.Boolean, _
    ByVal IsBlendClosed As System.Boolean, _
    ByVal KeepTangency As System.Boolean, _
    ByVal ForceNonRational As System.Boolean, _
    ByVal IsSolidBody As System.Boolean, _
    ByVal TessTolFactor As System.Double, _
    ByVal StartTangentLength As System.Double, _
    ByVal EndTangentLength As System.Double, _
    ByVal StartTangentDir As System.Boolean, _
    ByVal EndTangentDir As System.Boolean, _
    ByVal StartMatchingType As System.Integer, _
    ByVal EndMatchingType As System.Integer, _
    ByVal Merge As System.Boolean _
) As Body2

```

```

Dim instance As IModeler
Dim PModDoc As ModelDoc2
Dim Profile As System.Object
Dim GuideCurve As System.Object
Dim Centerline As System.Object
Dim IsThinBody As System.Boolean
Dim ThinType As System.Integer
Dim Thickness1 As System.Double
Dim Thickness2 As System.Double
Dim FeatureScope As System.Boolean
Dim IsBlendClosed As System.Boolean
Dim KeepTangency As System.Boolean
Dim ForceNonRational As System.Boolean
Dim IsSolidBody As System.Boolean
Dim TessTolFactor As System.Double
Dim StartTangentLength As System.Double
Dim EndTangentLength As System.Double
Dim StartTangentDir As System.Boolean
Dim EndTangentDir As System.Boolean
Dim StartMatchingType As System.Integer
Dim EndMatchingType As System.Integer
Dim Merge As System.Boolean
Dim value As Body2

```

```

value = instance.CreateLoftBody2(PModDoc, Profile, GuideCurve, Centerline, IsThinBody, ThinType,
Thickness1, Thickness2, FeatureScope, IsBlendClosed, KeepTangency, ForceNonRational, IsSolidBody,
TessTolFactor, StartTangentLength, EndTangentLength, StartTangentDir, EndTangentDir,
StartMatchingType, EndMatchingType, Merge)

```

```

Body2 CreateLoftBody2(
    ModelDoc2 PModDoc,
    System.object Profile,
    System.object GuideCurve,
    System.object Centerline,
    System.bool IsThinBody,
    System.int ThinType,
    System.double Thickness1,
    System.double Thickness2,
    System.bool FeatureScope,
    System.bool IsBlendClosed,

```

```

System.bool KeepTangency,
System.bool ForceNonRational,
System.bool IsSolidBody,
System.double TessTolFactor,
System.double StartTangentLength,
System.double EndTangentLength,
System.bool StartTangentDir,
System.bool EndTangentDir,
System.int StartMatchingType,
System.int EndMatchingType,
System.bool Merge
)

```

C++/CLI

```

Body2^ CreateLoftBody2(
& ModelDoc2^ PModDoc,
& System.Object^ Profile,
& System.Object^ GuideCurve,
& System.Object^ Centerline,
& System.bool IsThinBody,
& System.int ThinType,
& System.double Thickness1,
& System.double Thickness2,
& System.bool FeatureScope,
& System.bool IsBlendClosed,
& System.bool KeepTangency,
& System.bool ForceNonRational,
& System.bool IsSolidBody,
& System.double TessTolFactor,
& System.double StartTangentLength,
& System.double EndTangentLength,
& System.bool StartTangentDir,
& System.bool EndTangentDir,
& System.int StartMatchingType,
& System.int EndMatchingType,
& System.bool Merge
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

PModDoc

[Model document](#) in which to create the loft body

Profile

Array of sketches representing the profiles of the loft body

GuideCurve

Array of sketches representing the guide curves of the loft body

Centerline

Sketch representing a centerline of the loft body

IsThinBody

True if the loft body is to be a thin body, false if not

ThinType

- 0 = One direction
- 1 = One direction reverse
- 2 = Midplane
- 3 = Two direction

Thickness1

Value for thickness for first direction if a thin body

Thickness2

Value for thickness for second direction if a thin body

FeatureScope

True if the loft body only affects selected bodies, false if the loft body affects all bodies

IsBlendClosed

True for a closed loft, false for an open loft; if true and you selected less than three profiles, then any selected guide curves must be closed curves

KeepTangency

If the section curves are tangent, then you have the option to specify whether the resulting faces are also tangent; specify true to maintain the tangency as seen in the section curves, false to not

NOTE: When generating tangent surfaces, SOLIDWORKS maintains planar and cylindrical surface shapes if the section curves exhibit these characteristics.

ForceNonRational

True to force the resulting surface to be non-rational, false to not

IsSolidBody

True to return a solid loft body, false to return a surface loft body

TessTolFactor

Factor to control the number of intermediate sections used for loft with centerline; default value is 1.0; the greater the variable, the more intermediate sections created

StartTangentLength

Start tangent length

EndTangentLength

End tangent length

StartTangentDir

True is one direction, false is the opposite direction

EndTangentDir

True is one direction, false is the opposite direction

StartMatchingType

Tangency type at the start profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to the selected vector
- 3 = tangent to all adjacent faces sharing an edge with the start profile

EndMatchingType

Tangency type as the end profile:

- 0 = none
- 1 = tangent to the normal of the profile
- 2 = tangent to the selected vector
- 3 = tangent to all adjacent faces sharing an edge with the start profile

Merge

True merges the results in a multibody part, false does not

Return Value

Loft [body](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateLoftBody2](#).

Example

[Create Loft Surface Body and Change Into Feature \(VBA\)](#)

[Create Loft Body \(VB.NET\)](#)

[Create Loft Body \(VBA\)](#)

[Create Loft Body \(C#\)](#)

Remarks

Specifying guide curves and a centerline are optional. It is best to use guide curves, especially when you select profiles in the FeatureManager design tree.

You must specify the profiles in an order consistent with the desired direction of the loft. If creating a solid loft body, then the profiles must be closed. You can use any number of profiles. However, if you specify only one profile, then any specified guide curves must be closed curves.

Linear edge, sketch line, axis, plane and planar faces are qualified for tangency vector sections.

See Also

[IModeler Interface](#)

[IModeler Members](#)

[IFeatureManager::InsertCutBlend Method](#)

[IFeatureManager::InsertProtrusionBlend Method](#)

[IModelDoc2::InsertLoftRefSurface2 Method](#)

[IModeler::CreateLoftSurface Method](#)

[IModeler::ICreateLoftSurface Method](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

CreateLoftSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateLoftSurface Method (IModeler)

Creates a loft surface.

Visual Basic (Declaration)

```
Function CreateLoftSurface( _
    ByVal CurveArray As System.Object, _
    ByVal BBlendClosed As System.Boolean, _
    ByVal BForceCubic As System.Boolean, _
    ByVal GuideCrvArray As System.Object, _
    ByVal StartMatchingType As System.Integer, _
    ByVal EndMatchingType As System.Integer, _
    ByVal NormalAtStartSection As System.Object, _
    ByVal NormalAtEndSection As System.Object, _
    ByVal StartMatchingFaceList As System.Object, _
    ByVal EndMatchingFaceList As System.Object, _
    ByVal DegeneratedStart As System.Boolean, _
    ByVal DegeneratedEnd As System.Boolean, _
    ByVal StartPointOfStartSection As System.Object, _
    ByVal StartPointOfEndSection As System.Object, _
    ByVal SectionIndexStart As System.Integer, _
    ByVal SectionIndexEnd As System.Integer, _
    ByVal GuideIndexStart As System.Integer, _
    ByVal GuideIndexEnd As System.Integer _
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler
Dim CurveArray As System.Object
Dim BBlendClosed As System.Boolean
Dim BForceCubic As System.Boolean
Dim GuideCrvArray As System.Object
Dim StartMatchingType As System.Integer
Dim EndMatchingType As System.Integer
Dim NormalAtStartSection As System.Object
Dim NormalAtEndSection As System.Object
Dim StartMatchingFaceList As System.Object
Dim EndMatchingFaceList As System.Object
Dim DegeneratedStart As System.Boolean
Dim DegeneratedEnd As System.Boolean
Dim StartPointOfStartSection As System.Object
Dim StartPointOfEndSection As System.Object
Dim SectionIndexStart As System.Integer
Dim SectionIndexEnd As System.Integer
Dim GuideIndexStart As System.Integer
Dim GuideIndexEnd As System.Integer
Dim value As System.Object
```

```
value = instance.CreateLoftSurface(CurveArray, BBlendClosed, BForceCubic, GuideCrvArray,
StartMatchingType, EndMatchingType, NormalAtStartSection, NormalAtEndSection,
StartMatchingFaceList, EndMatchingFaceList, DegeneratedStart, DegeneratedEnd,
StartPointOfStartSection, StartPointOfEndSection, SectionIndexStart, SectionIndexEnd,
GuideIndexStart, GuideIndexEnd)
```

C#

```
System.object CreateLoftSurface(
    System.object CurveArray,
    System.bool BBlendClosed,
    System.bool BForceCubic,
    System.object GuideCrvArray,
    System.int StartMatchingType,
    System.int EndMatchingType,
    System.object NormalAtStartSection,
    System.object NormalAtEndSection,
    System.object StartMatchingFaceList,
    System.object EndMatchingFaceList,
    System.bool DegeneratedStart,
    System.bool DegeneratedEnd,
```

```

System.Object StartPointOfStartSection,
System.Object StartPointOfEndSection,
System.Int SectionIndexStart,
System.Int SectionIndexEnd,
System.Int GuideIndexStart,
System.Int GuideIndexEnd
)

```

C++/CLI

```

System.Object^ CreateLoftSurface(
& System.Object^ CurveArray,
& System.Bool BBlendClosed,
& System.Bool BForceCubic,
& System.Object^ GuideCrvArray,
& System.Int StartMatchingType,
& System.Int EndMatchingType,
& System.Object^ NormalAtStartSection,
& System.Object^ NormalAtEndSection,
& System.Object^ StartMatchingFaceList,
& System.Object^ EndMatchingFaceList,
& System.Bool DegeneratedStart,
& System.Bool DegeneratedEnd,
& System.Object^ StartPointOfStartSection,
& System.Object^ StartPointOfEndSection,
& System.Int SectionIndexStart,
& System.Int SectionIndexEnd,
& System.Int GuideIndexStart,
& System.Int GuideIndexEnd
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

CurveArray

Array of b-spline [curves](#)

BBlendClosed

True if blend closed, false if not

BForceCubic

True if force surface is cubic, false if not

GuideCrvArray

Array of guide [curves](#)

StartMatchingType

Start matching type:

- 0 = MATCH_NONE (default)
- 1 = MATCH_NORMAL
- 2 = MATCH_VECTOR
- 3 = MATCH_ALL_FACES or MATCH_FACE_G1
- 4 = MATCH_FACE_G2

EndMatchingType

End matching type:

- 0 = MATCH_NONE (default)
- 1 = MATCH_NORMAL
- 2 = MATCH_VECTOR
- 3 = MATCH_ALL_FACES or MATCH_FACE_G1
- 4 = MATCH_FACE_G2

NormalAtStartSection

[Normal at start section](#) when StartMatchingType is MATCH_NORMAL or MATCH_VECTOR; otherwise, can be Nothing

NormalAtEndSection

[Normal at start section](#) when EndMatchingType is MATCH_NORMAL or MATCH_VECTOR; otherwise, can be Nothing

StartMatchingFaceList

Array of matching [faces](#) at start section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be Nothing

EndMatchingFaceList

Array of matching [faces](#) at end section when startMatchingType is MATCH_ALL_FACES, MATCH_FACE_G1, or MATCH_FACE_G2; otherwise, can be Nothing

DegeneratedStart

True to degenerate at start, false to not

DegeneratedEnd

True to degenerate at end, false to not

StartPointOfStartSection

[Start point](#) of start section

StartPointOfEndSection

[Start point](#) of end section

SectionIndexStart

Index of start section; default is -1
SectionIndexEnd
Index of end section; default is -1
GuideIndexStart
Index of start guide curve; default is -1
GuideIndexEnd
Index of end guide curve; default is -1

Return Value

Newly created loft [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateLoftSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreateOffsetSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::CreateToroidalSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)

Availability

SOLIDWORKS 2006 FCS, Revision Numbers 14.0

SOLIDWORKS API Help

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CreateOffsetSurface Method (IModeler)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModeler Interface](#) : CreateOffsetSurface Method (IModeler)

Creates a surface that is offset from an existing surface.

.NET Syntax

Visual Basic (Declaration)

```
Function CreateOffsetSurface( _  
    ByVal SurfaceIn As System.Object, _  
    ByVal Distance As System.Double _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModeler  
Dim SurfaceIn As System.Object  
Dim Distance As System.Double  
Dim value As System.Object
```

```
value = instance.CreateOffsetSurface(SurfaceIn, Distance)
```

C#

```
System.Object CreateOffsetSurface(  
    System.Object SurfaceIn,  
    System.Double Distance  
)
```

C++/CLI

```
System.Object^ CreateOffsetSurface(  
& System.Object^ SurfaceIn,  
& System.Double Distance  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

SurfaceIn
Offset [surface](#)
Distance
Offset distance

Return Value

Newly created [surface](#)

Visual Basic for Applications (VBA) Syntax

See [Modeler::CreateOffsetSurface](#).

See Also

[IModeler Interface](#)
[IModeler Members](#)
[IBody2::CreateOffsetSurface Method](#)
[IBody2::ICreateOffsetSurface Method](#)
[IModeler::CreateBsplinesSurface Method](#)
[IModeler::CreateConicalSurface2 Method](#)
[IModeler::CreateCoonsBSurface Method](#)
[IModeler::CreateCylindricalSurface2 Method](#)
[IModeler::CreateExtrusionSurface Method](#)
[IModeler::CreateLoftSurface Method](#)
[IModeler::CreatePlanarSurface2 Method](#)
[IModeler::CreateRuledSurface Method](#)
[IModeler::CreateSphericalSurface2 Method](#)
[IModeler::CreateSweptSurface Method](#)
[IModeler::ICreateBsplinesSurface Method](#)
[IModeler::ICreateConicalSurface2 Method](#)
[IModeler::ICreateCylindricalSurface2 Method](#)
[IModeler::ICreateExtrusionSurface Method](#)
[IModeler::ICreateLoftSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)
[IModeler::ICreatePlanarSurface2 Method](#)
[IModeler::ICreateRuledSurface Method](#)
[IModeler::ICreateSphericalSurface2 Method](#)
[IModeler::ICreateSweptSurface Method](#)
[IModeler::ICreateToroidalSurface Method](#)
[IModeler::ICreateOffsetSurface Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0